

## UMA070 : ADVANCED NUMERICAL METHODS

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**Course Objective:** The main objective of this course is to motivate the students to understand and learn various advanced numerical techniques to solve mathematical problems governing various engineering and physical problems.

### Syllabus

**Non-Linear Equations:** Methods for multiple roots, Muller's, Iteration and Newton-Raphson method for non-linear system of equations and Newton-Raphson method for complex roots.

**Polynomial Equations:** Descartes' rule of sign, Birge-vieta, Giraffe's methods.

**System of Linear Equations:** Cholesky and Partition methods, SOR method with optimal relaxation parameters.

**Eigen-Values and Eigen-Vectors:** Similarity transformations, Gerschgorin's bound(s) on eigenvalues, Given's and Rutishauser methods.

**Interpolation and Approximation:** Cubic and B – Spline and bivariate interpolation, Least squares approximations, Gram-Schmidt orthogonalisation process and approximation by orthogonal polynomial, Legendre and Chebyshev polynomials and approximation.

**Differentiation and Integration:** Differentiation and integration using cubic splines, Romberg integration and multiple integrals.

**Ordinary differential Equations:** Milne's, Adams-Moulton and Adam's Bashforth methods with their convergence and stability, Shooting and finite difference methods for second order boundary value problems.

### Course Learning Objectives (CLO)

Upon completion of this course, the students will be able to:

1. find multiple roots of equation and apply Newton -Raphson's method to obtain complex roots as well solution of system of non - linear equations.
2. learn how to obtain numerical solution of polynomial equations using Birge - Vitae and Giraffe's methods.
3. apply Cholesky, Partition and SOR methods to solve system of linear equations.
4. understand how to approximate the functions using Spline, B- Spline, least square. approximations
5. learn how to solve definite integrals by using cubic spline, Romberg and initial value problems and boundary value problems numerically.

### Recommended Books:

1. Gerald, C.F. and Wheatley, P.O., Applied Numerical Analysis, Pearson Education (2008) 7<sup>th</sup> ed.
2. Gupta, S.R., Elements of Numerical Analysis, MacMillan India (2009).
3. Atkinson, K.E., An introduction to Numerical Analysis, John Wiley (2004) 2<sup>nd</sup> ed.
4. S.D. Conte, S.D. and Carl D. Boor, Elementary Numerical Analysis: An Algorithmic Approach, Tata McGraw Hill (2005).
5. Jain M. K., Iyengar. S.R.K. and Jain, R.K. Numerical Methods for Scientific and Engineering Computation, New Age International (2008) 5<sup>th</sup> ed